

Catching connected cars that lie about where they are

A signed message proves which car sent it, but it cannot prove that the position inside is true. In our simulations one receiver cannot detect a consistent lie, and receivers that pool their radio measurements catch most lies larger than about **39 m** (95 percent interval 34.5 to 45.5 m).

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1 The problem

Connected cars broadcast their position several times a second over a direct radio link, and the cars around them act on it. Every message is signed, so a car with valid credentials can still sign a false position, and no cryptographic check will reject it.

A receiver does have one independent clue: **how strongly the signal arrived**, which falls with distance from where the sender really is.

2 Aims and approach

- **Can one receiver tell** that a correctly signed position is false?
- **Can many receivers tell** by pooling what they measured, and from what size of lie?
- **Where would an informed attacker lie**, and what stops it?

We simulated a highway in the ns-3 network simulator [2, 3] with five traffic scenarios, ten kinds of attack, and honest cars that carry realistic position error. The result is a public dataset of **7,916,708 labelled one-second windows** [1]. Part A of this project reported a perfect score on a different dataset; an audit found 96.39 percent of its test rows were also in training, so the data was rebuilt and checked for leakage before any model was trained.

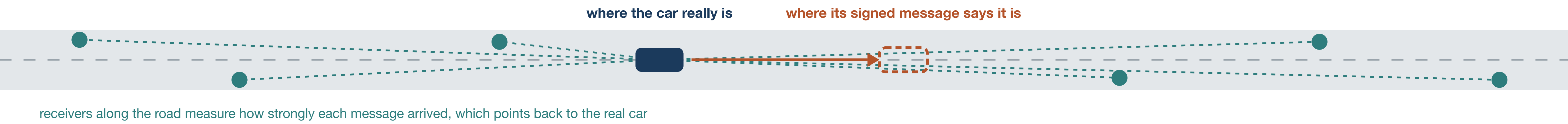


Figure 1. The attack studied here. The message is correctly signed, so only the physics of the radio signal can contradict the claim.

3 Result: a detection floor

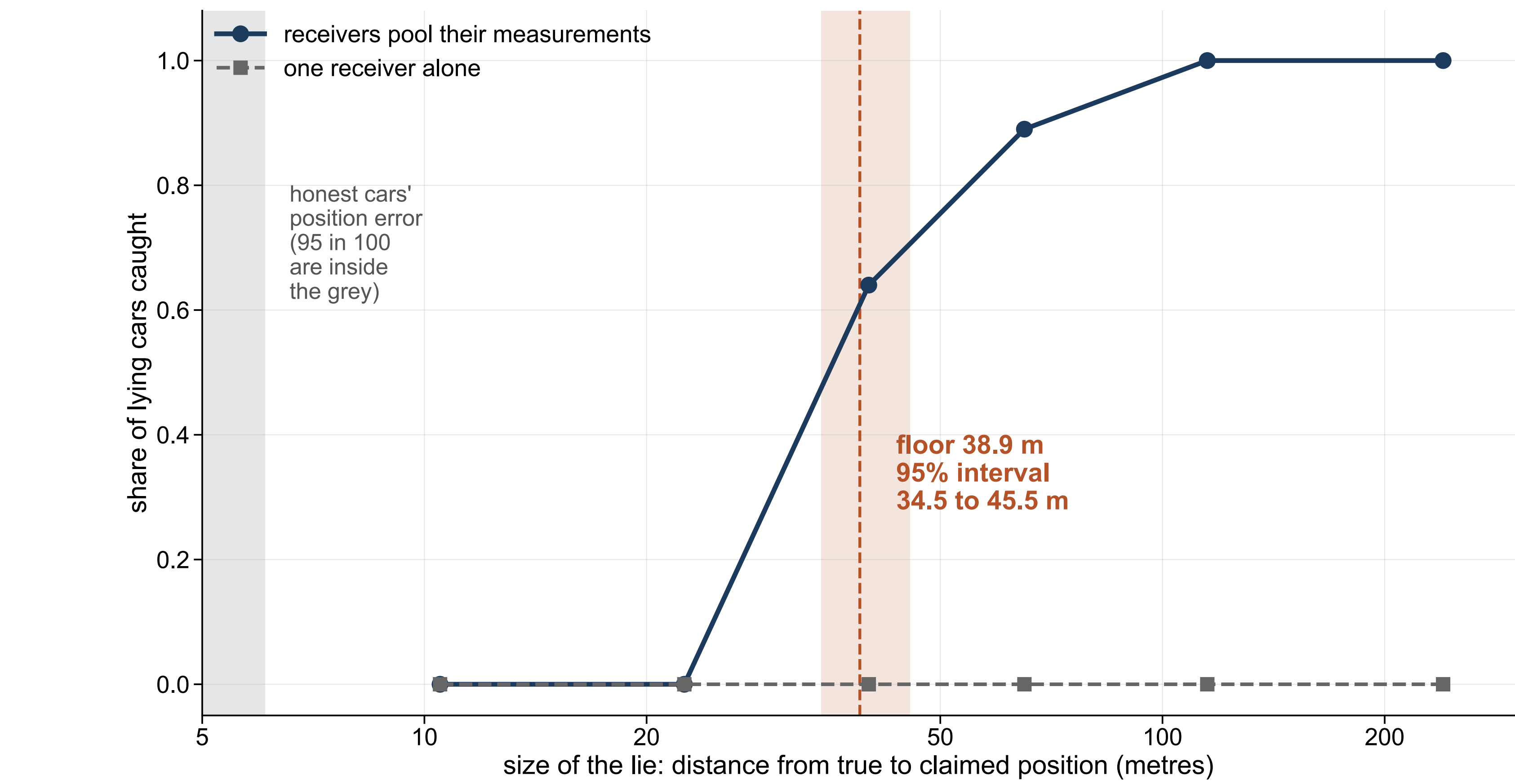


Figure 2. Pooling turns an undetectable lie into a detectable one above about twice the 18.3 m localisation error. Below the floor a lie is too small to separate from honest position error; a lie that small does not move a car into another lane. Some of the lies caught were also off the road, where a map check would reject them anyway (section 6).

One receiver sees nothing. A car that shifts every claim by the same amount keeps its speed and heading consistent. On a detection score from 0 (none caught) to 1 (all caught), content checks score 0.000 on it and four different learning methods reach at most 0.158.

0.000
detection score of content checks on a consistent lie

Pooling has to happen before the decision. When receivers vote with their verdicts instead of sharing measurements, detection of the same lies stays at exactly 0.000.

38.9 m
detection floor, 95 percent interval 34.5 to 45.5 m

The honest cars matter. Here they carry realistic position error. Without it every lie is separable in principle and the detection task becomes far easier than on a real road.

4 Result: the geometry has a weak side, and a map closes it

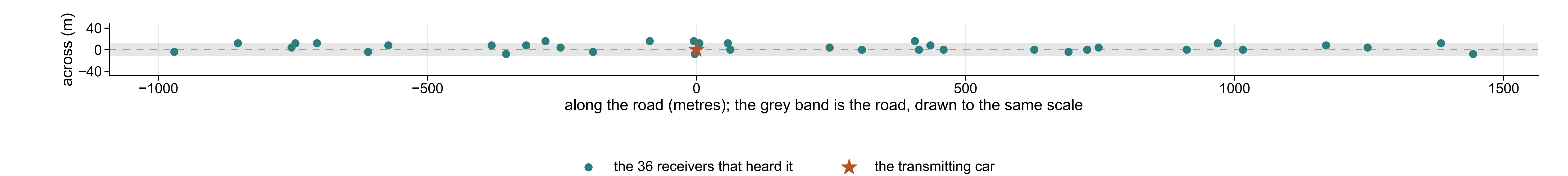


Figure 3. The receivers that heard one car span 2,414 m along the road and 24 m across it. They measure distance along the road well and across it poorly, so an attacker who knows this lies sideways.



Figure 4. Receiver positions alone predict the attacker's best direction: 79.7° off the road against 75 to 85° found by search, for the estimator this system runs.

5 Conclusions

1. **Aim 1.** A single receiver cannot detect a consistent position lie at any size in this data (Figure 2, flat line). The same holds on the public VeReMi datasets [4].
2. **Aim 2.** Pooling received power across receivers detects lies above a floor of 38.9 m, 95 percent interval 34.5 to 45.5 m (Figure 2). Sharing verdicts instead of measurements detects none.
3. **Aim 3.** Receiver geometry predicts where an informed attacker lies, for the estimator this system runs (Figures 3 and 4). A map check covers that direction (Table 1).

6 Limits and next step

All measurements are simulated, on one straight highway. Many simulated attackers claimed positions more than 18 m off the road (28.61 percent of the constant shift windows), which a map check rejects with no radio evidence. **Next: score pooled detection on on-road claims only**, to separate the radio's share of the floor from the map's.



Try it yourself: you are the attacker. Move a claimed position and watch the receivers decide.

References

1. L. L. Gowda, CV2X-IDS dataset, version 1.1.0, RMIT University, 2026, CC BY 4.0.
2. G. F. Riley and T. R. Henderson, "The ns-3 network simulator," Springer, 2010.
3. N. Patriciello et al., "An E2E simulator for 5G NR networks," Simul. Model. Pract. Theory, 2019.
4. R. W. van der Heijden, T. Lukaseder and F. Kargl, "VeReMi: A dataset for comparable evaluation of misbehavior detection," SecureComm, 2018.